Probability and Statistics 1
Chapter 2: Numerical representation of data

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1 Introduction

Chapter 1 produced, for a single character, a table and a few graphs. This is already a summary, but a table of ten rows and a histogram are still too much to compare, to transmit and to test. The present chapter goes one step further: it replaces the whole distribution by a small number of *parameters*, chosen so that each captures one aspect of the data.

Two families of parameters are used, and they answer two different questions. *Position parameters* answer the question *around which value is the distribution centred?*; they are the mode, the mean (in its several forms) and the median with its quartiles. *Dispersion parameters* answer the question *how far from that centre are the observations spread?*; they are the range, the mean absolute deviation, the variance and standard deviation, the coefficient of variation, and the interquartile range. A single number is never sufficient: a mean latency of 55 ms is compatible with a service that answers every request in 55 ms and with a service that answers half of the requests in 10 ms and half in 100 ms. Position alone describes a cloud by its centre of gravity; dispersion is what tells us whether the cloud is dense or scattered.

Example.

1. Two models obtain the same mean accuracy of 0.87 over ten training runs. The first is remarkably stable (standard deviation 0.002); the second oscillates between 0.81 and 0.93 (standard deviation 0.05). The mean is identical, the engineering decision is not: the second model cannot be deployed without a variance-reduction effort.

2. A service has a mean latency of 55 ms, but its 99th percentile is 220 ms. The mean describes the average user; the percentile describes the user who leaves. A report that gives only one of the two is incomplete, and often misleading.

A parameter is always a *loss of information*, and the choice of which parameter to compute is part of the statistical work, not an afterthought. The mean, the median, the geometric mean, the harmonic mean and the quadratic mean all answer the question "what is a typical value?" but they do so differently, and for some distributions their values differ enormously. The purpose of this chapter is to present each of them precisely, to show on examples when they coincide and when they diverge, and to state the properties that justify their use.

2 Position parameters

2.1 The mode

The mode is the simplest and the oldest of the position parameters, and the only one that applies to a nominal qualitative character.

Definition. Mode

The **mode** of a character is the modality (or the value, or the class) whose absolute frequency n_i is the largest. It is denoted Mo. The mode may fail to be unique; when it is unique it is called the *dominant* modality.

Example.

1. *Qualitative character.* In the survey on code editors of Chapter 1 (VS Code 38, Jupyter 18, Vim/Neovim 14, emacs 10), the mode is VS Code.
2. *Discrete quantitative character.* For the number of GPUs per node (1:12, 2:20, 4:16, 8:9, 16:3), the largest absolute frequency is 20, attained for 2 GPUs. So $Mo = 2$.
3. *Bimodal case.* In a survey on platforms (Discord 12, Instagram 12, TikTok 10, X 6), two modalities share the maximal frequency; the mode is the set {Discord, Instagram}.

For a continuous quantitative character grouped in classes, the mode is not a class but a number, obtained by an interpolation inside the modal class. The interpolation weights the position of the mode inside the class by the excess of the modal frequency over its two neighbours.

Definition. Mode of a continuous variable

Let $[a_i, b_i[$ be the *modal class*, of amplitude $a = b_i - a_i$, of frequency n_i. Let n_{i-1} and n_{i+1} be the frequencies of the adjacent classes, and put $e_1 = n_i - n_{i-1}$, $e_2 = n_i - n_{i+1}$. The **mode** is the number

$$Mo = a_i + \frac{e_1}{e_1 + e_2}\, a.$$

Example.

For the inference latency, the modal class is $[50; 60[$ with $n_i = 41$, preceded by $n_{i-1} = 22$ and followed by $n_{i+1} = 21$. We have $e_1 = 19$, $e_2 = 20$, $a = 10$, hence

$$Mo = 50 + \frac{19}{19 + 20} \times 10 = 50 + 4.87 \approx 54.87 \text{ ms}.$$

The mode is close to the centre of the modal class because the two adjacent classes have almost the same frequency.

Remark.

The mode is the only position parameter available for a nominal character: computing a mean or a median requires additions or an order, neither of which exists on labels. It is also the only parameter that can be read directly on the bar chart or the histogram, since it corresponds to the highest bar. Its drawback is its instability: a small change in the sample can move the mode from one modality to another when the two highest frequencies are close.

2.2 The arithmetic mean

The mean is the most used position parameter, and also the one whose properties are the most thoroughly exploited in the rest of the course.

Definition. Arithmetic mean

Let $x_1, x_2, \dots, x_k$ be the k modalities of a quantitative character X, observed with absolute frequencies $n_1, \dots, n_k$ on a sample of size $n = \sum_i n_i$. The **arithmetic mean** of X is the number

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{k} n_i x_i.$$

Example.

1. *Accuracy across runs.* A model is trained ten times with different seeds and yields

$$0.865,\ 0.871,\ 0.869,\ 0.874,\ 0.870,\ 0.868,\ 0.872,\ 0.869,\ 0.873,\ 0.869.$$

Here $n = 10$ and each $n_i = 1$, so the mean is

$$\bar{X} = \frac{1}{10} \sum_{i=1}^{10} x_i = 0.870.$$

2. *GPUs per node.* the mean is

$$\bar{X} = \frac{1}{60}\big(12 \cdot 1 + 20 \cdot 2 + 16 \cdot 4 + 9 \cdot 8 + 3 \cdot 16\big) = \frac{236}{60} \approx 3.93,$$

where the values $1, 2, 4, 8, 16$ are the modalities and $12, 20, 16, 9, 3$ their frequencies.

When the observations do not carry the same importance, each is given a weight, and the mean becomes a weighted mean.

Definition. Weighted arithmetic mean

Let $x_1, \dots, x_k$ be numbers and $w_1, \dots, w_k$ positive weights. The **weighted mean** of the x_i with weights w_i is

$$\bar{X}_w = \frac{\sum_{i=1}^{k} w_i x_i}{\sum_{i=1}^{k} w_i}.$$

When the weights are the absolute frequencies, $\bar{X}_w$ coincides with the arithmetic mean of the frequency table.

Example.

1. *Overall accuracy on an imbalanced dataset.* A three-class classifier reaches 0.95 accuracy on class 1 (1000 examples), 0.80 on class 2 (100 examples) and 0.60 on class 3 (100 examples). The unweighted mean of the three accuracies is 0.783, but the true overall accuracy is the weighted mean

$$\bar{X}_w = \frac{1000 \cdot 0.95 + 100 \cdot 0.80 + 100 \cdot 0.60}{1200} = \frac{1090}{1200} \approx 0.908.$$

The unweighted mean overweights the two minority classes and is not the accuracy of the model on the dataset.

2. *Mean latency of a service.* Two endpoints are served, `/health` with 1000 requests per minute at 5 ms and `/search` with 100 requests per minute at 200 ms. The unweighted mean of the two latencies is 102.5 ms, but the mean latency over all requests is

$$\bar{X}_w = \frac{1000 \cdot 5 + 100 \cdot 200}{1100} = \frac{25000}{1100} \approx 22.7 \text{ ms}.$$

The difference is a factor of four: the mean latency of a service is always weighted by its traffic, never computed endpoint by endpoint.

Three elementary properties of the mean are used constantly.

Proposition. *Properties of the arithmetic mean*

Let X be a quantitative character, $\bar{X}$ its arithmetic mean, and a, b two real numbers. Then:

(i) *Balance.* $\sum_{i=1}^{k} n_i (x_i - \bar{X}) = 0$: the sum of the deviations from the mean is zero.

(ii) *Linearity.* If $Y = aX + b$, then $\bar{Y} = a\bar{X} + b$.

(iii) *Combination.* If a sample of size n_1 has mean $\bar{X}_1$ and a disjoint sample of size n_2 has mean $\bar{X}_2$, the mean of their union is

$$\bar{X} = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}.$$

The first property is the one that justifies the name: the mean is the point around which the positive and negative deviations compensate exactly. The second says that a change of origin or of unit propagates without surprise. The third is used each time a dataset is split into subgroups, for instance when computing the mean latency of a fleet from the mean latency of each region.

Remark.

The arithmetic mean is *sensitive to extreme values*. A single observation far from the others can displace it considerably; e.g. for the latency data , the mean is 54.9 ms, while the median, computed below, is 54.88 ms; on this roughly symmetric distribution the two coincide. On a distribution with a long right tail, on the other hand, the mean is pulled towards the tail and exceeds the median, sometimes by a large factor. This asymmetry is the reason why service-level objectives are stated in percentiles, not in means.

2.3 The geometric, harmonic and quadratic means

The arithmetic mean is not the only mean, and for some quantities it is the wrong one. Ratios, rates and errors are typical cases: the natural average of a ratio is not its arithmetic mean.

Definition. Geometric mean

Let $x_1, \dots, x_n$ be n strictly positive numbers. The **geometric mean** is the number

$$G = \left(x_1 x_2 \cdots x_n\right)^{1/n} = \exp\left(\frac{1}{n} \sum_{i=1}^{n} \ln x_i\right).$$

If the values appear with frequencies n_i, the geometric mean of the frequency table is $G = \left(\prod_{i=1}^{k} x_i^{n_i}\right)^{1/n}$ with $n = \sum_i n_i$.

Example.

A compilation pass is applied to five programs, and yields speedups of 1.5, 1.8, 2.0, 1.6 and 1.4 with respect to the original code. The geometric mean of the speedups is

$$G = (1.5 \cdot 1.8 \cdot 2.0 \cdot 1.6 \cdot 1.4)^{1/5} = (12.096)^{1/5} \approx 1.646,$$

while the arithmetic mean is $(1.5 + 1.8 + 2.0 + 1.6 + 1.4)/5 = 1.66$. The geometric mean is the standard way of summarising speedup ratios in benchmark suites, precisely because it is the only mean that treats a speedup of $2\times$ and a slowdown of $0.5\times$ symmetrically.

Definition. Harmonic mean

Let $x_1, \ldots, x_n$ be n strictly positive numbers. The **harmonic mean** is the number

$$H = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}.$$

If the values appear with frequencies n_i, $H = \dfrac{n}{\sum_{i=1}^{k} n_i / x_i}$.

Example.

The *F1 score* of an information-retrieval system, the quantity reported next to precision and recall in every classification report, is the harmonic mean of precision P and recall R:

$$F_1 = \frac{2}{\frac{1}{P} + \frac{1}{R}} = \frac{2PR}{P + R}.$$

For $P = 0.8$ and $R = 0.6$, the harmonic mean is $0.96/1.4 \approx 0.686$, while the arithmetic mean would be 0.70. The harmonic mean penalises the imbalance between the two, which is precisely the property wanted: a system with very high precision and very low recall is useless, and the F1 score says so.

Definition. Quadratic mean (root mean square)

Let $x_1, \ldots, x_n$ be n real numbers. The **quadratic mean**, or root mean square, is the number

$$Q = \sqrt{\frac{1}{n} \sum_{i=1}^{n} x_i^2}.$$

If the values appear with frequencies n_i, $Q = \sqrt{\dfrac{1}{n} \sum_{i=1}^{k} n_i x_i^2}$.

Example.

A regression model is evaluated on five test points, on which its errors are -0.01, $+0.02$, -0.015, $+0.005$ and -0.005. The root mean square error is

$$Q = \sqrt{\frac{0.0001 + 0.0004 + 0.000225 + 0.000025 + 0.000025}{5}} = \sqrt{0.000155} \approx 0.0124.$$

The mean of the absolute errors is 0.011, slightly smaller: the RMS error penalises large errors more than small ones, which is why it is the standard loss of regression.

The four means are ordered, and the inequality that relates them is one of the most useful facts of the theory.

Proposition. *Chain of means*

For strictly positive numbers, the four means satisfy

$$H \leq G \leq \bar{X} \leq Q,$$

with equality in each comparison if and only if all the values are equal.

The inequality is rarely used directly, but it explains the examples: on the speedups, $H \approx 1.61 < G \approx 1.646 < \bar{X} = 1.66$; on the F1 example, $H = 0.686 < \bar{X} = 0.70$. Choosing the mean is therefore choosing *where* in this chain the summary of the data should sit, and the choice depends on the question.

Remark.

The arithmetic mean is the mean of *sums*: if quantities add, their means add. The geometric mean is the mean of *products* and ratios: if quantities multiply (growth rates, speedups, relative errors), their geometric mean is the natural summary. The harmonic mean is the mean of *rates*: if a quantity is the ratio of a total to a total (throughput, F1 score), its average is a harmonic mean. The quadratic mean is the mean of *squares*: it is used each time the squared error is the loss, which is the case in least-squares regression. Using the arithmetic mean for a rate is a classical error; it is not a matter of notation but of meaning.

2.4 The median

The median is the position parameter that resists extreme values. Unlike the mean, it is defined for an ordinal qualitative character, since it only requires the modalities to be ordered.

Definition. Median

Let X be a quantitative or ordinal qualitative character, and let $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$ be the observations of a sample of size n sorted in increasing order. The **median**, denoted Me, is the number

$$Me = \begin{cases} x_{((n+1)/2)} & \text{if } n \text{ is odd,} \\ \dfrac{x_{(n/2)} + x_{(n/2+1)}}{2} & \text{if } n \text{ is even.} \end{cases}$$

It divides the sample into two parts of equal size: at most half of the observations are below Me, at most half are above.

Example.

1. *Odd sample.* The eleven accuracy values $0.865, 0.868, 0.869, 0.869, 0.869, 0.870, 0.871, 0.872, 0.873, 0.874, 0.880$ give, for $n = 11$, $Me = x_{(6)} = 0.870$, the same as the mean up to the third decimal.

2. *Even sample.* The ten accuracies of the previous subsection, sorted, are $0.865, 0.868, 0.869, 0.869, 0.869, 0.870, 0.87$ the median is $(0.869 + 0.870)/2 = 0.8695$.

3. *Discrete quantitative character.* For the GPUs per node example, $n = 60$ is even, and the cumulative frequencies show that the 30th and 31st observations both equal 2; the median is 2. The mean was 3.93: the median is closer to the bulk of the data, which is concentrated on small values, while the mean is pulled upward by the few nodes with 8 or 16 GPUs.

For a continuous quantitative character grouped in classes, the median is obtained by linear interpolation inside the median class, that is, the class whose cumulative frequency reaches $n/2$.

Definition. Median of a continuous variable

Let $[a_i, b_i[$ be the *median class*, n_i its frequency, N_{i-1} the cumulative frequency of the preceding class and $a = b_i - a_i$ its amplitude. The **median** is

$$Me = a_i + \frac{\dfrac{n}{2} - N_{i-1}}{n_i}\, a.$$

Example.

For the latency data example, $n/2 = 50$; the cumulative frequency reaches 30 at the end of $[40; 50[$ and 71 at the end of $[50; 60[$, so the median class is $[50; 60[$, with $a_i = 50$, $n_i = 41$, $N_{i-1} = 30$, $a = 10$. Hence

$$Me = 50 + \frac{50 - 30}{41} \times 10 = 50 + 4.88 \approx 54.88 \text{ ms}.$$

This is very close to the mean of the grouped data, 54.9 ms: the latency distribution is almost symmetric.

The interpolation formula is nothing other than Thales' theorem applied to the cumulative frequency curve. Between a_i and b_i, the cumulative frequency is assumed to grow linearly from N_{i-1} to $N_{i-1}+n_i$; the value $n/2$ is reached at the abscissa given by the formula.

Proposition. *Properties of the median*

Let Me be the median of a quantitative character. Then:

(i) Me is insensitive to *any change of the observations above or below it*: replacing the largest observation by a much larger one does not modify Me.

(ii) If $Y = aX + b$ with $a > 0$, then $Me_Y = a\, Me_X + b$.

(iii) For a symmetric distribution (histogram symmetric about a vertical axis), the median and the mean coincide.

The first property is the reason why the median is called a *robust* parameter. A single abnormal latency, a single mislabeled point in a dataset, a single corrupted log entry can displace the mean substantially, but it cannot displace the median beyond one rank in the ordered series.

2.5 The quartiles and the interquartile range

The median divides the sample into two halves. The quartiles divide it into four parts of equal size, and describe the dispersion of the central half of the data.

Definition. Quartiles

The **first quartile** Q_1, the **second quartile** Q_2 and the **third quartile** Q_3 are the three values that divide the ordered sample into four parts containing the same number of observations. By construction, $Q_2 = Me$.

For a discrete variable, the computation of Q_1 and Q_3 follows the same rule as for the median, applied to the two halves of the sample.

Example.

Consider the 21 latencies, sorted, $4, 5, 6, 7, 7, 8, 9, 10, 10, 10, 11, 11, 11, 12, 12, 13, 13, 15, 15, 16, 17$. Since $n = 21 = 2 \times 10 + 1$, the median is $Q_2 = x_{(11)} = 10$. The first half contains the ten values below the median; it has even size $10 = 2 \times 5$, so $Q_1 = (x_{(5)} + x_{(6)})/2 = (7 + 8)/2 = 7.5$. The second half contains the ten values above the median; similarly $Q_3 = (x_{(16)} + x_{(17)})/2 = (12 + 13)/2 = 12.5$.

For a continuous variable grouped in classes, the quartiles are obtained by interpolation, exactly as the median, replacing $n/2$ by $n/4$ and $3n/4$.

Definition. Quartiles of a continuous variable

Let $[a_i, b_i[$ be the class containing the fraction $n/4$ (for Q_1) or $3n/4$ (for Q_3) of the observations, n_i its frequency and N_{i-1} the cumulative frequency of the preceding class. Then

$$Q_1 = a_i + \frac{\frac{n}{4} - N_{i-1}}{n_i}\, a, \qquad Q_3 = a_i + \frac{\frac{3n}{4} - N_{i-1}}{n_i}\, a.$$

Example.

For the latency data example, $n/4 = 25$: the cumulative frequency reaches 8 at the end of $[30; 40[$ and 30 at the end of $[40; 50[$, so Q_1 lies in $[40; 50[$ and

$$Q_1 = 40 + \frac{25 - 8}{22} \times 10 = 40 + 7.73 \approx 47.73 \text{ ms}.$$

Similarly $3n/4 = 75$, and the cumulative frequency reaches 71 at the end of $[60; 70[$ and 92 at the end of $[60; 70[$ itself, so Q_3 lies in $[60; 70[$ and

$$Q_3 = 60 + \frac{75 - 71}{21} \times 10 = 60 + 1.90 \approx 61.90 \text{ ms}.$$

The median was 54.88 ms, so half of the requests are served between 47.73 and 61.90 ms.

Definition. Interquartile range

The **interquartile range** is the difference between the third and the first quartile:

$$IQR = Q_3 - Q_1.$$

It is the length of the interval containing the central 50 % of the observations.

Example.

1. On the latency data, $IQR = 61.90 - 47.73 \approx 14.17$ ms: the central half of the requests is spread over 14 ms, whereas the range of the whole sample is 50 ms. The IQR is therefore about a quarter of the range, which indicates a moderately concentrated distribution.
2. On the 21 latencies above, $IQR = 12.5 - 7.5 = 5$, to be compared with the range $17 - 4 = 13$. Here the central half is even tighter relative to the total extent.

Remark.

The quartiles and the IQR are the robust counterparts of the standard deviation and of the range: they do not use the numerical values of the observations beyond their ranks, so they are insensitive to outliers. This is why the service-level objectives of a production system are stated as percentiles (p_{50} is the median, p_{25} and p_{75} are the quartiles, p_{99} is the value below which 99 % of the requests are served), and not as a mean and a standard deviation. The quartiles are percentiles, and the IQR is the interval $[p_{25}, p_{75}]$.

3 Dispersion parameters

Two samples may share the same mean, the same median and the same mode, and be entirely different. A service that answers every request in 55 ms and a service that answers half of the requests in 10 ms and half in 100 ms have the same mean latency, and they are not the same service. Dispersion parameters measure this difference: they quantify how far the observations lie from their centre. Each of them uses a different notion of distance, and each has its domain of validity.

3.1 The range

Definition. Range

The **range** of a quantitative character is the difference between the largest and the smallest observation:

$$r = x_{\max} - x_{\min}.$$

Example.

1. The latency data span the interval from 30 ms to 80 ms (in terms of class boundaries), hence $r = 50$ ms.
2. The accuracy values across the ten runs lie between 0.865 and 0.874, hence $r = 0.009$: on the scale of accuracies, this is a very small range.

The range is the simplest dispersion parameter, and the least informative. It depends on only two observations, the two most extreme ones, and is therefore destroyed by a single outlier. It is useful as an indicator of the total extent of the data, for instance to choose the bounds of a histogram, but it is not a measure of dispersion in the sense of the next parameters.

3.2 The mean absolute deviation

The natural way to measure dispersion is to compute the distance of each observation from the centre, and to average these distances. The first average that comes to mind uses absolute values.

Definition. Mean absolute deviation

Let $x_1, \dots, x_n$ be the observations of a quantitative character and $\bar{X}$ their mean. The **mean absolute deviation** is the number

$$e_X = \frac{1}{n} \sum_{i=1}^{n} \left| x_i - \bar{X} \right|.$$

If the values appear with frequencies n_i, $e_X = \dfrac{1}{n} \displaystyle\sum_{i=1}^{k} n_i \left| x_i - \bar{X} \right|$.

Example.

On the ten accuracies of the previous subsection, the deviations from the mean $\bar{X} = 0.870$ are $0.005, 0.001, 0.001, 0.004, 0.000, 0.002, 0.002, 0.001, 0.003, 0.001$ in absolute value. Their sum is 0.020, and the mean absolute deviation is $e_X = 0.002$. The interpretation is direct: on average, a training run is off by two thousandths of accuracy from the mean.

The mean absolute deviation is expressed in the same unit as the character, which makes it interpretable. It has nevertheless a major defect: the absolute value has no simple algebraic expression, so e_X cannot be decomposed into components, cannot be differentiated, and does not combine linearly when samples are merged. The next parameter replaces the absolute value by a square, which restores all the algebraic properties.

3.3 The variance and the standard deviation

Definition. Variance

Let $x_1, \dots, x_k$ be the modalities of a quantitative character X, observed with frequencies $n_1, \dots, n_k$ on a sample of size n. The **variance** of X is the number

$$\operatorname{Var}(X) = \sigma_X^2 = \frac{1}{n} \sum_{i=1}^{k} n_i \left(x_i - \bar{X} \right)^2.$$

Definition. Standard deviation

The **standard deviation** of X is the square root of the variance:

$$\sigma_X = \sqrt{\operatorname{Var}(X)}.$$

Example.

1. *Accuracy across runs.* Using the deviations already computed, the sum of their squares is 0.000062, and
$$\sigma_X^2 = \frac{0.000062}{10} = 6.2 \times 10^{-6}, \qquad \sigma_X \approx 0.0025.$$

The mean accuracy is 0.870 and the standard deviation is 0.0025: the model is stable at the scale of the second decimal.

2. *GPUs per node.* the mean is $\bar{X} \approx 3.93$. The sum of squared deviations, computed with the formula below, gives
$$\sigma_X^2 \approx 28.2 - 3.93^2 \approx 12.7, \qquad \sigma_X \approx 3.57.$$
The standard deviation is of the same order as the mean, which announces a very heterogeneous fleet: some nodes have one GPU, others sixteen.

The direct computation of the variance requires two passes over the data, one for the mean and one for the sum of squared deviations. A single-pass formula, known as König's formula, is preferred in practice.

Proposition. *König's formula*

$$\operatorname{Var}(X) = \frac{1}{n}\sum_{i=1}^{k} n_i x_i^2 - \bar{X}^{\,2} = \overline{x^2} - (\bar{x})^2.$$

The formula is obtained by expanding $(x_i - \bar{X})^2 = x_i^2 - 2\bar{X}x_i + \bar{X}^2$ and using $\sum_i n_i x_i = n\bar{X}$. It shows that the variance is the mean of the squares minus the square of the mean, i.e. that the second moment of the distribution carries the dispersion. It is valid for any quantitative character, and it extends to the grouped case by replacing x_i by the class centre.

Example.

For the latency data, the class centres are $35, 45, 55, 65, 75$ with frequencies $8, 22, 41, 21, 8$. The mean of the squares is

$$\overline{x^2} = \frac{8\cdot 35^2 + 22\cdot 45^2 + 41\cdot 55^2 + 21\cdot 65^2 + 8\cdot 75^2}{100} = \frac{312100}{100} = 3121,$$

and the mean is $\bar{X} = 54.9$. König's formula gives

$$\sigma_X^2 = 3121 - 54.9^2 \approx 3121 - 3014.0 = 107.0, \qquad \sigma_X \approx 10.34 \text{ ms}.$$

Two properties of the variance are used constantly in the rest of the course and in any statistical computation.

Proposition. *Properties of the variance*

Let X be a quantitative character and a, b two real numbers. Then:

(i) $\operatorname{Var}(X) \geq 0$, with equality if and only if all the observations are equal.

(ii) *Invariance by translation:* $\operatorname{Var}(X + b) = \operatorname{Var}(X)$.

(iii) *Homogeneity of degree two:* $\operatorname{Var}(aX) = a^2 \operatorname{Var}(X)$, hence $\sigma_{aX+b} = |a|\,\sigma_X$.

(iv) *Combination of disjoint samples:*

$$\operatorname{Var}(X) = \frac{n_1 \operatorname{Var}(X_1) + n_2 \operatorname{Var}(X_2)}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2}\big(\bar{X}_1 - \bar{X}_2\big)^2.$$

Property (iii) is the one that justifies the use of the standard deviation rather than the variance in reports: the standard deviation is in the same unit as the character, and it transforms like a length under a change of unit. Property (iv), sometimes called the formula of total variance, is the seed of the analysis of variance and of the decomposition of the variance of a bivariate series in Chapter 3.

Remark.

The variance is *not robust*. Since each deviation is squared, an observation at distance 10 contributes 100 times more than an observation at distance 1; a single outlier can double the variance. When the data contain extreme values, or when the distribution has a heavy tail, the mean absolute deviation or the IQR should be reported alongside the standard deviation, and the median alongside the mean. A report that gives only a mean and a standard deviation on a heavy-tailed distribution is not wrong, but it describes a different dataset than the one observed.

3.4 The coefficient of variation

The standard deviation is expressed in the unit of the character, which makes it directly interpretable but also impossible to compare across characters measured in different units. The coefficient of variation removes the unit.

Definition. Coefficient of variation

Let X be a quantitative character with mean $\bar{X} > 0$ and standard deviation σ_X. The **coefficient of variation** of X is the dimensionless number

$$CV_X = \frac{\sigma_X}{\bar{X}},$$

often expressed as a percentage, $CV_X = 100\,\sigma_X / \bar{X}$.

Example.

1. *Latency.* $\bar{X} = 54.9$ ms and $\sigma_X = 10.34$ ms give $CV_X \approx 0.188$, that is $18.8\,\%$. The dispersion is moderate.
2. *Accuracy.* $\bar{X} = 0.870$ and $\sigma_X = 0.0025$ give $CV_X \approx 0.0029$, that is $0.29\,\%$. The dispersion is negligible. On the same scale, the latency and the accuracy cannot be compared by their standard deviations (10.34 versus 0.0025), but their coefficients of variation say that the latency is about sixty times more dispersed, relatively, than the accuracy.
3. *GPUs per node.* $\bar{X} = 3.93$ and $\sigma_X = 3.57$ give $CV_X \approx 0.908$, that is $90.8\,\%$. The fleet is very heterogeneous.

Proposition. *Comparison of dispersions*

The coefficient of variation is invariant under a change of unit and of origin, in the sense that $CV_{aX+b} = CV_X$ for any $a > 0$ and any b. It therefore allows the comparison of the dispersion of two characters, whatever their units.

By convention, a sample with $CV_X \le 0.25$ is said to be *homogeneous*, a sample with $0.25 < CV_X \le 0.80$ *moderately scattered*, and a sample with $CV_X > 0.80$ *very heterogeneous*. These thresholds are

conventions, not theorems; their purpose is to give a common vocabulary to reports and dashboards, and they should be stated when they are used.

Remark.

The coefficient of variation requires $\bar{X} > 0$, which excludes characters centred on zero or taking negative values (returns, errors, temperatures in degrees Celsius). It is also meaningless for a character whose zero is not absolute: metrics like temperature or relative score differences have arbitrary zeros, so the CV of such a metric should be interpreted with care, whereas a latency has an absolute zero and its CV is fully interpretable.

3.5 The boxplot and the outliers

The parameters above summarise the distribution by a few numbers. The **boxplot** summarises it visually, by placing on a single axis the five numbers $x_{\min}$, Q_1, Me, Q_3, $x_{\max}$, and by flagging the observations that lie far from the box.

Definition. Boxplot

A **boxplot** of a quantitative character is constructed as follows.

1. Draw a rectangle whose lower side is Q_1 and upper side is Q_3, and inside it a line at the median Me.
2. Compute the *inner fences* $L = Q_1 - 1.5\,IQR$ and $U = Q_3 + 1.5\,IQR$.
3. Let x_ℓ be the smallest observation still above L, and x_h the largest observation still below U. Draw a segment (the *lower whisker*) from Q_1 to x_ℓ and a segment (the *upper whisker*) from Q_3 to x_h.
4. Any observation strictly below x_ℓ or strictly above x_h is an **outlier**, drawn as an isolated point.

The rule $1.5\,IQR$ is a convention introduced by John Tukey in 1977. It marks as outliers the observations that lie far enough from the box to deserve individual attention; it does not mean that they are erroneous, only that they are atypical.

Example.

Three API endpoints are monitored, with 20 latency measurements each (in ms):

- *Health check:* $5, 6, 6, 7, 7, 7, 8, 8, 8, 8, 9, 9, 9, 10, 10, 11, 12, 12, 13, 15$;
- *Search:* $20, 22, 25, 28, 30, 32, 35, 38, 40, 42, 45, 48, 50, 55, 60, 65, 70, 80, 90, 120$;
- *Recommendation:* $30, 32, 33, 34, 35, 35, 36, 36, 37, 37, 38, 38, 39, 39, 40, 42, 45, 48, 55, 90$.

The five-number summaries are:

	$x_{\min}$	Q_1	Me	Q_3	$x_{\max}$
Health check	5	7	8.5	10.5	15
Search	20	31	43.5	57.5	120
Recommendation	30	35	37.5	39.5	90

For the health check, $IQR = 3.5$, $L = 1.75$ and $U = 15.75$: the whiskers extend from 5 to 15, and there is no outlier. For the search, $IQR = 26.5$, $L = -8.75$ and $U = 97.25$: the upper whisker ends at 90, and 120 is an outlier. For the recommendation, $IQR = 4.5$, $L = 28.25$ and $U = 46.25$: the whiskers extend from 30 to 45, and both 55 and 90 are outliers. The three boxplots are shown in Figure 3.1.

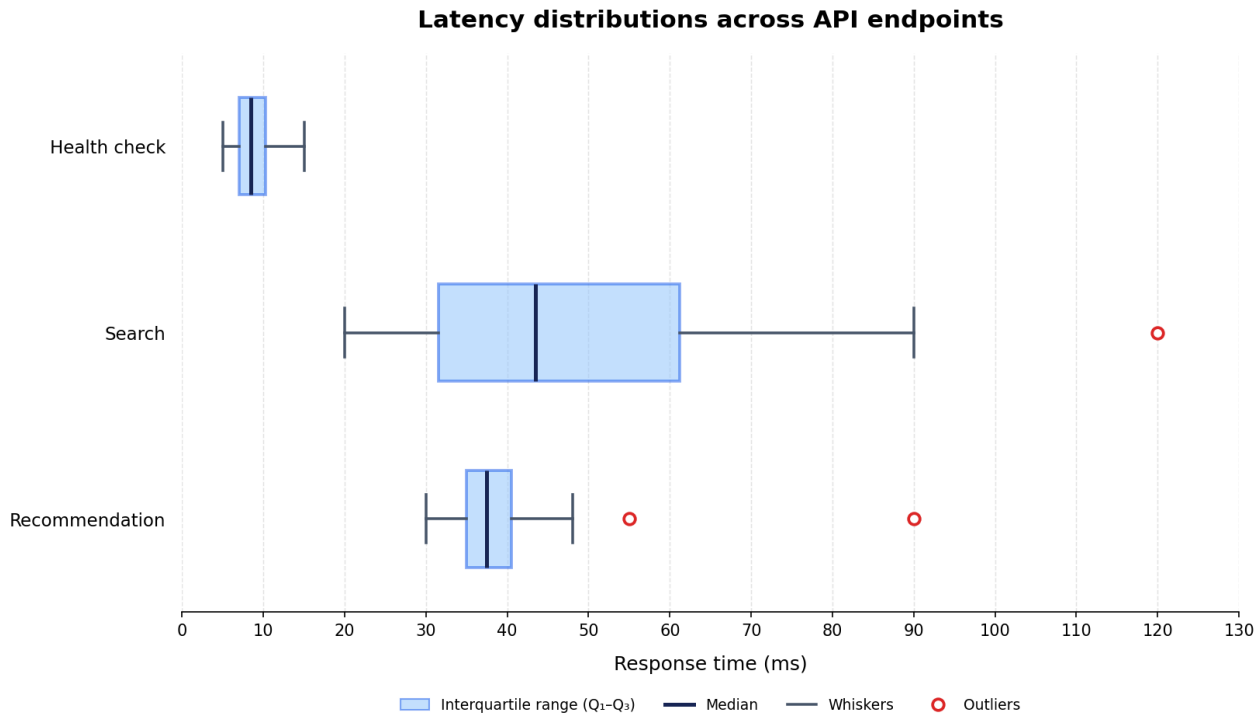


Figure 3.1: Boxplots of the latencies of three API endpoints. The box goes from Q_1 to Q_3, the line inside is the median, the whiskers extend to the most extreme observations within $1.5\,IQR$ of the box, and the isolated points are outliers.

Remark.

The boxplot is the only representation in this chapter that superimposes position, dispersion, asymmetry and outliers on a single axis. Its asymmetry is read in the position of the median inside the box: a median close to Q_1 announces a distribution stretched towards the upper values, a median close to Q_3 the opposite. For the recommendation endpoint, the median is almost at the centre of the box (37.5 against a box from 35 to 39.5), so the bulk of the data is symmetric, but the two outliers on the right show that a small fraction of the requests is much slower than the rest: this is the typical signature of a heavy-tailed service, invisible on the mean and on the standard deviation taken alone.

4 Which parameter for which question?

The parameters of this chapter are not interchangeable. Each answers a specific question, and using the wrong one is not a numerical error but an error of interpretation. The following table summarises the domains of validity.

Parameter	Type of character	Robustness	Use
Mode	all (incl. nominal)	insensitive to order	dominant modality
Median	ordinal, quantitative	robust to outliers	central value of skewed data
Arithmetic mean	quantitative	sensitive to outliers	balance point, linear operations
Weighted mean	quantitative	sensitive to outliers	aggregation with unequal importance
Geometric mean	positive quantitative (ratios)	sensitive to outliers	growth rates, speedup benchmarks
Harmonic mean	positive quantitative (rates)	sensitive to outliers	average rate, F1 score
Quadratic mean (RMS)	quantitative	very sensitive to outliers	squared error, RMSE
Range	quantitative	very sensitive to a single value	full extent, choice of classes
Mean absolute deviation	quantitative	robust	typical deviation, same unit as X
Variance / std. dev.	quantitative	sensitive to outliers	inference, variance decomposition
Coefficient of variation	positive quantitative	sensitive to outliers	comparison across units or scales
Quartiles, IQR	ordinal, quantitative	robust to outliers	central spread, service-level objectives

Table 4.1: Domain of validity of the position and dispersion parameters.

A few practical rules follow from the table.

- For a *symmetric* distribution without outliers, the mean, the median and the mode are close, and any of them describes the centre. The standard deviation is then the natural dispersion.
- For a *skewed* distribution (latency, income, file sizes), the median and the IQR are the honest summary; the mean and the standard deviation are still computed, but they must be reported next to the median, not instead of it.
- For a *nominal* character, only the mode and the frequency distribution are available.
- For a *ratio* or a *rate*, the geometric or the harmonic mean is the correct average; the arithmetic mean is not, even when the computation is possible.
- For a *comparison* across characters measured in different units, the coefficient of variation is the only dimensionless dispersion available.

5 Common mistakes

1. **Confusing mean and median on a skewed distribution.** A mean latency of 55 ms on a heavy-tailed service says nothing about the typical user. The median is the value to report, and the mean should be accompanied by it, not replace it.

2. **Using the arithmetic mean for a rate.** The average throughput of two machines that process 10 and 1000 requests per second is not $(10 + 1000)/2 = 505$, it is the harmonic mean $2/(1/10 + 1/1000) \approx 19.8$. The arithmetic mean overestimates the average rate whenever the rates are unequal.

3. **Using the arithmetic mean for a ratio of ratios.** Speedups, relative errors and growth rates are averaged geometrically. The arithmetic mean of $2\times$ and $0.5\times$ is 1.25, which suggests an overall speedup, whereas the true combined factor is 1 (they cancel exactly).

4. **Forgetting to weight.** The mean accuracy of a model on an imbalanced dataset is the weighted mean of the class accuracies, weighted by the class sizes. The unweighted mean overweights the minority classes and does not measure the overall accuracy.

5. **Computing a variance on one observation.** The variance of a sample of size one is zero by the formula, but it is meaningless: a variance requires at least two observations, and a reliable variance requires many more.

6. **Comparing standard deviations across units.** A standard deviation of 10 is large for an accuracy (scale 0 to 1) and small for a latency in milliseconds. To compare dispersions across units, use the coefficient of variation, and only if the mean is positive and the zero of the scale is meaningful.

7. **Reading a boxplot as a histogram.** The box of a boxplot contains 50 % of the observations, but its area is not proportional to that fraction. Two boxplots with the same width may hide very different distributions; the boxplot complements the histogram, it does not replace it.

8. **Treating outliers as errors.** An outlier is an atypical observation, not necessarily an erroneous one. The Tukey rule flags it for inspection; deleting it without examination is a form of data manipulation.

6 Summary

- *Position parameters* locate the centre of the distribution. The *mode* is the most frequent value (or class); it is the only one available for a nominal character. The *arithmetic mean* is $\bar{X} = \frac{1}{n}\sum_i n_i x_i$, generalised to the *weighted mean* $\bar{X}_w = \sum_i w_i x_i / \sum_i w_i$. The *geometric*, *harmonic* and *quadratic* means average products, rates and squares respectively, and they satisfy $H \le G \le \bar{X} \le Q$. The *median* Me divides the sample into two halves, and the *quartiles* $Q_1, Q_2 = Me, Q_3$ divide it into four parts; for a continuous variable they are computed by interpolation inside the median class.

- *Dispersion parameters* quantify the spread around the centre. The *range* is $x_{\max} - x_{\min}$; the *mean absolute deviation* is $\frac{1}{n}\sum_i n_i |x_i - \bar{X}|$; the *variance* is $\sigma_X^2 = \frac{1}{n}\sum_i n_i (x_i - \bar{X})^2$, computed in one pass by König's formula $\sigma_X^2 = \overline{x^2} - \bar{X}^2$, and the *standard deviation* σ_X is its square root, in the unit of the character; the *coefficient of variation* $CV_X = \sigma_X / \bar{X}$ is the dimensionless dispersion used to compare characters measured in different units.

- The *interquartile range* $IQR = Q_3 - Q_1$ is the robust counterpart of the standard deviation. The *boxplot* superimposes $x_{\min}, Q_1, Me, Q_3, x_{\max}$ and flags as outliers the observations beyond $Q_1 - 1.5\,IQR$ or $Q_3 + 1.5\,IQR$.

- No parameter is universally correct. Symmetric data without outliers are correctly described by the mean and the standard deviation; skewed or heavy-tailed data require the median and the IQR; ratios and rates require the geometric and harmonic means; comparisons across scales require the coefficient of variation.
- Chapter 3 extends this apparatus to two characters observed simultaneously: covariance, linear correlation coefficient, conditional means and variances, and the regression line by least squares. The variance decomposition of the present chapter (property (iv) of Proposition 3.3) is the seed of that extension.

References

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[3] J. L. Devore, *Probability and Statistics for Engineering and the Sciences*, 9th edition, Cengage, 2016. Chapter 1 treats the same parameters, with an emphasis on the comparison between mean and median on skewed data.

[4] G. Saporta, *Probabilités, analyse des données et statistique*, 3rd edition, Technip, 2011. Chapters 1–2 are the French reference for the terminology used here (*moyenne harmonique*, *moyenne géométrique*, *écart absolu moyen*, *étendue interquartile*).

[5] J. W. Tukey, *Exploratory Data Analysis*, Addison-Wesley, 1977. The origin of the boxplot and of the $1.5\,IQR$ rule.

[6] M. P. Deisenroth, A. A. Faisal, C. S. Ong, *Mathematics for Machine Learning*, Cambridge University Press, 2020. Chapter 6 gives the probabilistic reading of the mean and the variance, and Sections 6.4–6.5 the variance decomposition used in Chapter 3.

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